



CarthaBot — Program over Wi-Fi (Full Setup)

Set the robot up once, then code wirelessly on your own Wi-Fi — every step, in detail.

CarthaBot Companion · ESP32-C3 bridge · in-app Wi-Fi provisioning

The big picture — 3 one-time setups, then code forever:

Part A put MicroPython on the robot (the app does it over USB). **Part B** flash the Wi-Fi bridge firmware onto the ESP32-C3 (once, Arduino IDE). **Part C** tell the robot your Wi-Fi *inside the app*, over the USB cable (type name + password → Save — no network switching). Then **Part D**: code wirelessly on your own network, internet kept, no cables. Each part below is fully detailed.

PART A Put MicroPython on the robot (one time, USB)

This makes the robot understand the programs the app sends. The app does it automatically — you only need a USB cable once.

- 1 Plug the **CarthaBot** into the PC with a **USB-C cable** and turn it on.
- 2 Open the **CarthaBot Companion** app and open any coding activity (Kids / under-6 VPL / Advanced Python).
- 3 Choose **USB** when asked. The app flashes MicroPython and shows "**CarthaBot is ready.**"
- 4 That's it. The app **also** auto-installs a tiny `boot.py` on the robot so it can be reached wirelessly later — nothing for you to do.

You only repeat Part A if you ever re-flash the robot or change its firmware. Normal day-to-day use never needs it again.

PART B Flash the Wi-Fi bridge onto the ESP32-C3 (one time)

The small **DFRobot Beetle ESP32-C3** wired inside the robot is what carries your programs over Wi-Fi. Flash it once with the provisioning firmware so the app can set its Wi-Fi later.

B1 · Install the tools (first time only)

- 1 Install the **Arduino IDE** (1.8.x or 2.x) from arduino.cc.
- 2 In Arduino IDE: **File** ▶ **Preferences** ▶ **Additional Boards Manager URLs** and add:
`https://espressif.github.io/arduino-esp32/package\_esp32\_index.json`
- 3 **Tools** ▶ **Board** ▶ **Boards Manager...**, search **esp32**, install "**esp32 by Espressif**".

- 4 **Tools** ▶ **Manage Libraries...**, search **NimBLE-Arduino**, install it. (If your network blocks it, download the ZIP and use **Sketch** ▶ **Include Library** ▶ **Add .ZIP Library**.)

B2 · Open and flash the firmware

- 1 Plug the **Beetle ESP32-C3** into the PC by its **own USB-C** port (separate from the robot).

- 2 Open this sketch in Arduino IDE:

```
Downloads\CarthaBotCompanion-  
Junior\Firmware\esp32c3_bridge_v4_appwifi\esp32c3_bridge_v4_appwifi.ino
```

- 3 **Tools** ▶ **Board** → select "**ESP32C3 Dev Module**" (or your Beetle entry).

- 4 **Tools** ▶ **Port** → pick the COM port that appeared when you plugged the Beetle in.

- 5 **Tools** ▶ **USB CDC On Boot** ▶ **Enabled** (lets you see status messages while testing).

- 6 Click **Upload** (→ arrow). Wait for "**Done uploading.**"

- 7 Open **Tools** ▶ **Serial Monitor** at **115200**. On reset you should see lines like `[wifi]`
`AP=192.168.4.1` .

- 8 Unplug the Beetle from the PC and make sure it's wired back into the robot (powered from the robot's 3V3).

Wiring reminder (only if you ever rewire): Pico `GP0` → ESP `GPI04` , ESP `GPI05` → Pico `GP1` (CN1 SCL), and 3V3/GND shared. Cross the two signal wires.

PART C

Connect CarthaBot to *your* Wi-Fi — in the app (one time)

You do this once, **over the USB cable** — no CarthaBot-WiFi, no network switching, your PC stays on your normal Wi-Fi the whole time. The robot remembers your Wi-Fi and rejoins it on every power-up.

- 1 Plug the **CarthaBot** into the PC with the **USB-C cable** and turn it on.
- 2 Open the **CarthaBot Companion** app and open any coding activity.
- 3 In the connection panel, find the **WiFi** card and click **⚙️ First time? Put the robot on your Wi-Fi**.
- 4 Type your **Wi-Fi name (SSID)** and **password** in the two boxes.
- 5 Click **💾 Save to robot (USB cable)**. Wait for the green message: **"Sent! The robot is joining '<your wifi>' ... Robot address: carthabot.local:3333."**
- 6 Click **Close** and **unplug the USB cable**. Give the robot ~10 seconds to join your Wi-Fi.

That's the one-time setup finished. The robot is now on your own Wi-Fi and rejoins it automatically every power-up. You never touch CarthaBot-WiFi, and your PC never leaves your normal network.

PART D

Code wirelessly, every day (no cables, internet kept)

- 1 Turn on the robot. Make sure your PC is on **your normal Wi-Fi** (the same one you gave the robot).
- 2 Open the app → open a coding activity (Kids / under-6 VPL / Advanced Python).
- 3 In the connection panel pick **WiFi**. The address is already `carthabot.local:3333`.
- 4 Press **Connect** → the app finds the robot on your network automatically (it tries `carthabot.local`, then quickly scans your network if needed) → **"CarthaBot is ready."**
- 5 Build a program and press **▶ Play** — the robot runs it instantly, over the air. 🎨

You don't need to know the robot's IP — pressing **Connect** auto-discovers it on your network (this also covers the case where Windows can't resolve `carthabot.local`). If you prefer, you can still type the robot's IP (e.g. `192.168.1.xx:3333`) in the address box.

Alternative: Bluetooth (BLE) — zero setup, keeps internet

Don't want to deal with Wi-Fi at all? In the connection panel choose **Bluetooth (BLE)** instead. It pairs to a device named **CarthaBot**, keeps your PC's internet, and needs no network setup — just slightly slower for big programs. (Works after Parts A & B.)

Troubleshooting

Problem	Fix
"Couldn't find the robot on USB"	Plug the robot in with the USB cable and turn it on, and make sure you flashed the v4 firmware (Part B) and did Part A.
"CarthaBot couldn't join" after entering password	Re-check the Wi-Fi password (case-sensitive). 5 GHz-only networks won't work — the ESP32-C3 is 2.4 GHz; pick your 2.4 GHz network.
Connected but Play does nothing	The robot needs MicroPython + <code>boot.py</code> (Part A — the app installs both over USB). Do Part A once.
<code>carthabot.local</code> won't connect	Just press Connect — the app auto-scans your network for the robot. If still not found, confirm the robot is on your 2.4 GHz Wi-Fi and powered on.
Want to change the robot to a different Wi-Fi	Repeat Part C over USB: open the setup screen, type the new network name + password, Save.
Connection "sticks" / won't reconnect	The v4 firmware fixes this (a new connection always wins). If on older firmware, power-cycle the robot.

Quick reference

- **Setup:** done in-app over the USB cable — type your Wi-Fi name + password, press Save
- **After setup:** robot at `carthabot.local:3333` on your own Wi-Fi
- **Fallback AP:** `CarthaBot-WiFi` / `carthabot` · `192.168.4.1` (only if it can't join your Wi-Fi)
- **BLE name:** `CarthaBot`
- **Firmware to flash:** `Firmware\esp32c3_bridge_v4_appwifi\esp32c3_bridge_v4_appwifi.ino`
- **Wiring:** Pico GP0→ESP GPIO4 · ESP GPIO5→Pico GP1 (CN1 SCL) · 3V3 & GND shared

CarthaBot Companion — Wi-Fi programming setup. Parts A & B are one-time per robot; Part C is one-time per Wi-Fi network and is done over the USB cable, so your PC never leaves your normal network. The ESP32-C3 is 2.4 GHz only — pick your 2.4 GHz network.